CSE 111 Fall 2026: Project Checkpoint 1

ModStack

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Synopsis

Game Mod Tracker is a database application that organizes and manages information about video games and their mods. The system helps users find and track mods by storing details such as mod versions, categories, developers, supported platforms, download counts, and compatibility with different game versions. The main users (Gamers, Mod Developers, and Administrators) interact with the database to search for mods, manage mod releases, update compatibility information, and leave ratings and reviews. The database also enables users to generate reports on the most popular mods, games with the most mods, and commonly used mod categories.

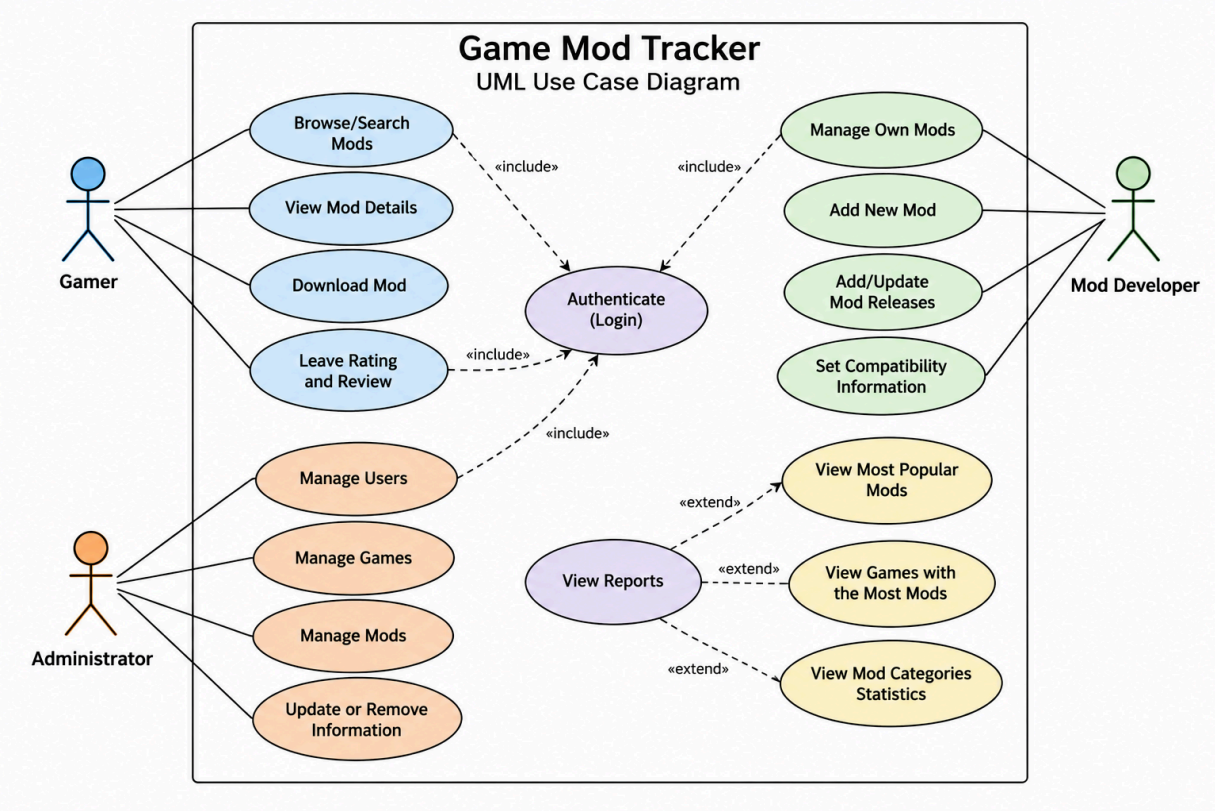
Authors & Contribution: 50% Alizeh, 50% Kevin

UML Use Case Diagram

The primary use case of the Game Mod Tracker is searching for mods and checking their compatibility with different game versions. In this workflow, gamers can search for mods based on a specific game or category. Once a user selects a mod and game version, the system checks whether the available mod releases are compatible with that game version. Users can also view mod details, access download links, and leave ratings and reviews to help others find mods. Mod Developers can add new mods, update mod information, manage different mod releases, and update compatibility records. Administrators are responsible for managing users, games, game versions, and mod information to keep the database organized and accurate. Secondary use cases include viewing popular mods, finding games with the most mods, and generating reports on different mod categories.

Authors & Contribution: 0% Alizeh, 100% Kevin

Use Case Diagram:

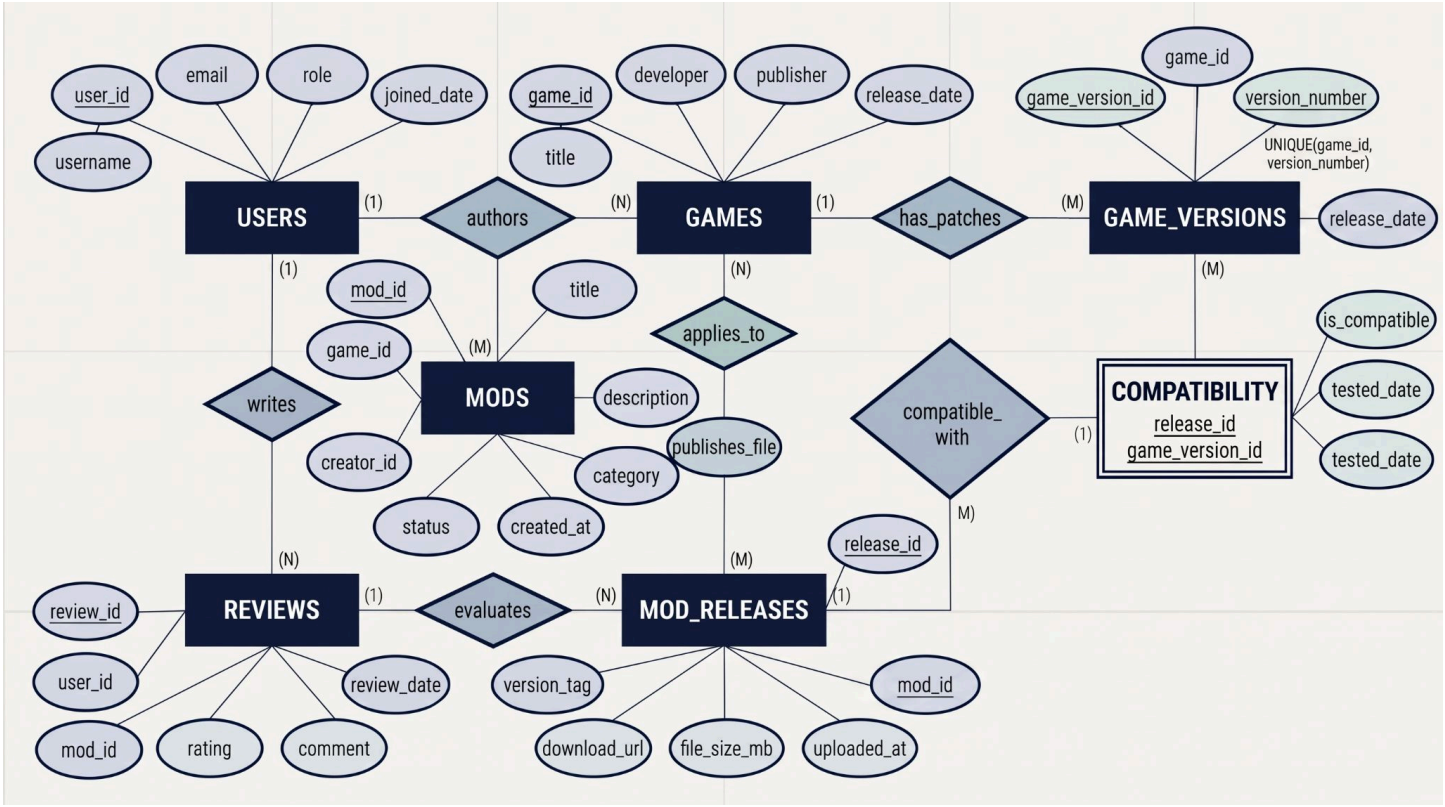


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ER Diagram

Our ER diagram contains seven entities: Users, Games, Game_Versions, Mods, Mod_Releases, Compatibility, and Reviews. It shows how these entities are connected through relationships, including one-to-many and many-to-many relationships. Compatibility and Reviews connect related entities and store additional information.

ERD-1:



Authors & Contribution: 100% Alizeh, 0% Kevin

Relational Schema

The database uses seven relations to manage users, games, game versions, mods, mod releases, compatibility, and reviews. Each relation has a primary key to uniquely identify each record. Foreign keys connect related relations, such as linking mods to their creators and games. The Compatibility relation uses a composite primary key consisting of release_id and game_version_id to identify each mod release and game version pairing. The Reviews relation connects users and mods, allowing users to rate and comment on mods.

1. Users

Users(user_id, username, email, role, joined_date)

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2. Games

Games(game_id, title, developer, publisher, release_date)

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3. Game_Versions

Game_Versions(game_version_id, game_id, version_number, release_date)

Foreign Key: game_id references Games(game_id)

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4. Mods

Mods(mod_id, game_id, creator_id, title, description, category, status, created_at)

Foreign Keys:

game_id references Games(game_id)

creator_id references Users(user_id)

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5. Mod_Releases

Mod_Releases(release_id, mod_id, version_tag, download_url, file_size_mb, download_count, uploaded_at)

Foreign Key: mod_id references Mods(mod_id)

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6. Compatibility

Compatibility(release_id, game_version_id, is_compatible, tested_date)

Composite Primary Key: (release_id, game_version_id)

Foreign Keys:

release_id references Mod_Releases(release_id)

game_version_id references Game_Versions(game_version_id)

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7. Reviews

Reviews(review_id, user_id, mod_id, rating, comment, review_date)

Primary Key: review_id

Foreign Keys:

user_id references Users(user_id)

mod_id references Mods(mod_id)

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